

GROUP IT, MAP IT! SETS & VENN DIAGRAMS

Grade 5 Mathematics Curriculum Companion

Topic: Set Theory

Grade Level: 5th Grade

Subject: Mathematics

1. Ready for the Picnic!

Our Story Begins...

Ananya and Kabir are packing for their school picnic to **Kanha National Park**. They have lots of snacks, games, and supplies, but everything is completely jumbled up! Can you help them sort and group their things? Let's get packing!

Sorting Indian Street Food!

Help Ananya and Kabir sort their favorite Indian street foods:

List of Snacks: Samosa, Murukku, Pani Puri, Banana Chips, Pakora, Mirchi Bajji, Kolhapuri Misal.

Think & Sort: Which snacks are *CRISPY*? Which are *SPICY*? Can some snacks be **BOTH**?

2. What is a Set?

In mathematics, a **Set** is a collection of well-defined objects that belong together based on a common rule or property.

Key Rule: You know exactly what belongs in a set and what does not!

Set Name	Elements in the Set	Common Property / Rule
Rohan's Geometry Box	{Triangle, Circle, Square, Rectangle, Ruler}	Tools used for geometry.
Traditional Indian Breakfast	{Idli, Dosa, Poha, Paratha, Upma}	Popular Indian breakfast items.
South Indian States	{Tamil Nadu, Kerala, Karnataka, Andhra Pradesh, Telangana}	The 5 states located in South India.

3. Overlapping Sets & Venn Diagrams

A **Venn Diagram** uses overlapping circles to visually organize items and show relationships between different sets.

Case Study: Crispy vs. Spicy Snacks 🌶️ Crunch!

- **Crispy Snacks Set:** {Samosa, Murukku, Banana Chips, Kolhapuri Misal}
- **Spicy Snacks Set:** {Mirchi Bajji, Kolhapuri Misal}
- **Both Crispy & Spicy (Overlap):** {Pani Puri, Pakora}

4. Key Operations: Union & Intersection

Concept	Definition	Example from Snacks
Union of Sets ($\cup$)	ALL items in Set A OR Set B (or both). Combines everything into one group.	{Samosa, Banana Chips, Murukku, Pani Puri, Pakora, Mirchi Bajji, Kolhapuri Misal}
Intersection of Sets ($\cap$)	Items that belong to BOTH Set A AND Set B at the same time (the overlap region).	{Pani Puri, Pakora}

5. Advanced Set Concepts

Disjoint Sets (No Overlap) 🚫

Some sets have **NO common elements**—they don't overlap at all! These are called **Disjoint Sets**.

- **Railway Vehicles:** {Train, Metro, Tram, Monorail}
- **Flying Vehicles:** {Aeroplane, Helicopter, Hot Air Balloon}

Since no vehicle can be both a railway vehicle AND a flying vehicle, these two circles never touch!

The Universal Set 🌐

The **Universal Set** includes ALL objects being considered in a specific problem or context.

Example: All 30 students in Class 5A love ice cream!

- **Universal Set:** All 30 students in Class 5A.
- **Subsets:** Mango Kulfi lovers and Chocolate Ice Cream lovers.
- **Overlap:** Students who love BOTH Mango Kulfi and Chocolate!

6. Problem-Solving with Venn Diagrams

Example Problem: Dogs Chasing Balls & Cats 🐶

Problem Statement: In a park, 12 dogs chase balls, 10 dogs chase cats, and 4 dogs chase both. How many dogs chase ONLY balls?

Step-by-Step Solution:

- Total dogs chasing balls = **12**
- Dogs chasing BOTH = **4**
- Dogs chasing ONLY balls = Total chasing balls – Dogs chasing both
- **$12 - 4 = 8$ dogs**

Answer: Exactly **8 dogs** chase only balls!

Group It, Map It! Practice Worksheet

Grade 5 - Sets & Venn Diagrams

Student Name: _____

Date: _____

Score: _____ / 20

Question 1: Definition of a Set

Which of the following represents a well-defined set?

- A) A collection of the most delicious snacks in India.
- B) The collection of states in South India {Tamil Nadu, Kerala, Karnataka, Andhra Pradesh, Telangana}.
- C) A group of very tall students in Class 5.
- D) A collection of fun games to play at a picnic.

Question 2: Understanding Venn Diagrams

In a Venn diagram showing two overlapping circles (Set A: Kids who play Cricket, Set B: Kids who play Badminton), what does the middle overlapping region represent?

Question 3: Disjoint Sets Identification

Explain why the set of *Railway Vehicles* {Train, Metro, Tram} and the set of *Flying Vehicles* {Aeroplane, Helicopter, Hot Air Balloon} are called **Disjoint Sets**. Draw a simple sketch showing how their circles would look in a Venn diagram.

Question 4: Word Problem — Sports in the Colony

In a neighborhood colony, 15 children play Cricket, 12 children play Badminton, and 5 children play BOTH Cricket and Badminton.

- a) How many children play **ONLY** Cricket?
- b) How many children play **ONLY** Badminton?
- c) What is the total number of children who play at least one of these sports (the Union of both sets)?

Question 5: Universal Set & Ice Cream Survey

In a class of 30 students (Universal Set), 18 students like Mango Kulfi, 15 like Chocolate Ice Cream, and 7 like BOTH. How many students like **ONLY** Mango Kulfi?